

Database Fundamentals & Design

Dorothy Guirgues
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Data & Information

- Data is raw, unorganized facts that need to be processed. Data can be something simple and seemingly random and useless until it is organized.
- When data is processed, organized, structured or presented in a given context so as to make it useful, it is called information.

Data & Information (cont.)

➤ Examples of Data and Information:

- The history of temperature readings all over the world for the past 100 years is data. If this data is organized and analyzed to find that global temperature is rising, then that is information.
- The number of visitors to a website by country is an example of data. Finding out that traffic from the U.S. is increasing while that from Australia is decreasing is meaningful information.

What is a database?

- “A database is a collection of information that is organized so that it can be easily accessed, managed and updated.”
- The data is typically organized to model relevant aspects of reality in a way that supports processes requiring this information.



Why Database?

- Storing data in a text file .
- Storing data in a spreadsheet .



Advantages of Database

- Redundancy can be reduced.
- Inconsistency can be avoided.
- Data can be shared.
- Security restrictions can be applied.
- Enforcing Integrity Constraints.
- Providing Backup and Recovery.

Disadvantages of Database

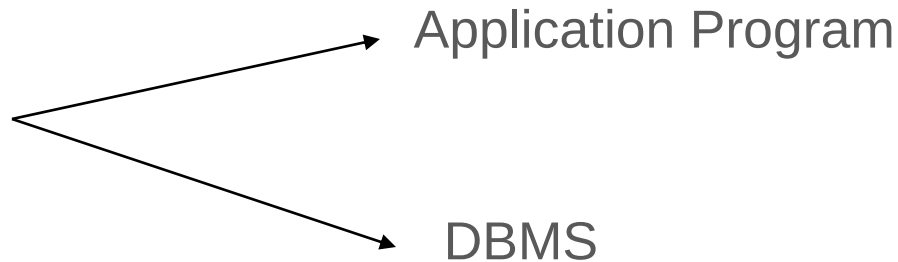
- It needs expertise to use (which is expensive).
- DBMS itself is expensive.
- DBMS may be incompatible with any other available DBMS.

Database System

➤ A database system is composed of :

✓ The database.

✓ The Software.



Types Of Database

- Centralized database.
- Distributed database.
- Personal database.
- End-user database.
- Commercial database.
- NoSQL database.
- Operational database.

Types Of Database (Cont.)

- Relational database.
- Cloud database.
- Object-oriented database.
- Graph database.

For more info:

<https://www.tutorialspoint.com/Types-of-databases>

Database Management System (DBMS)

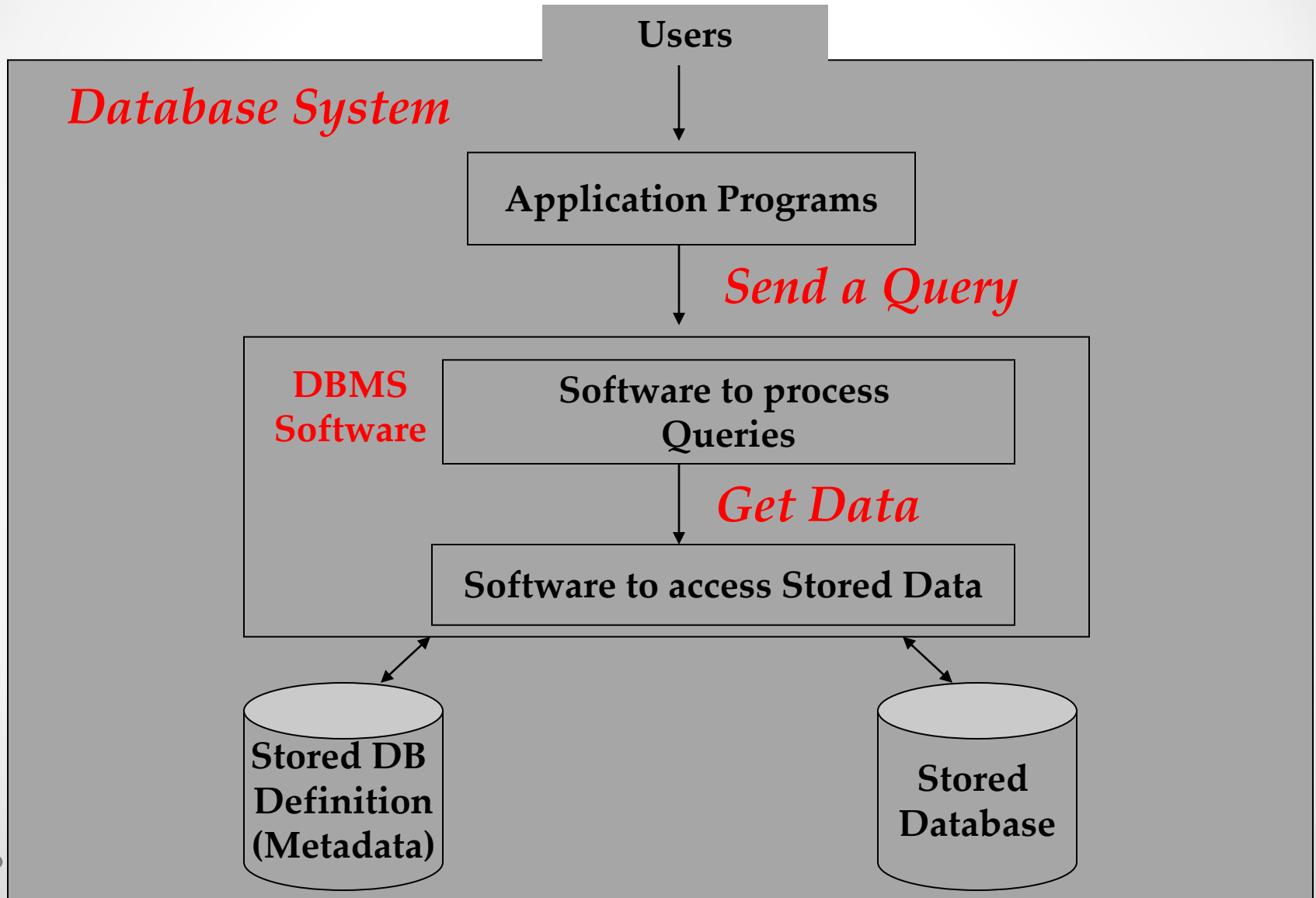
- It is system software for creating and managing databases. It provides users and programmers with a systematic way to create, retrieve, update and manage data.
- It essentially serves as an interface between the database and end users or application programs, ensuring that data is consistently organized and remains easily accessible.



DBMS Functions

- Defining Database.
- Constructing Database.
- Manipulating Database.
- Data Security & Integrity.
- Concurrency.
- Backup & Recovery.
- Data Dictionary (Meta Data).
- Performance.

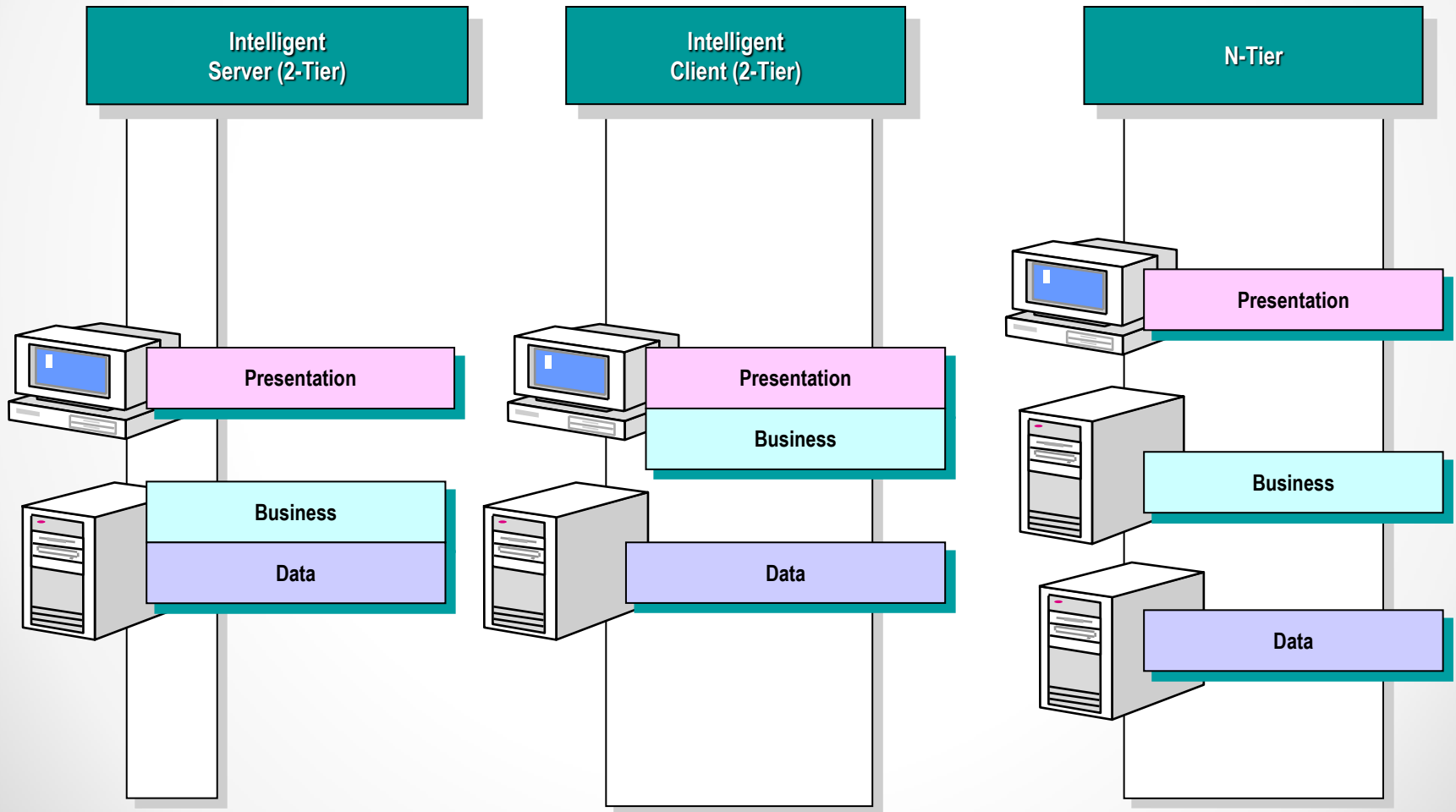
Database System (cont.)



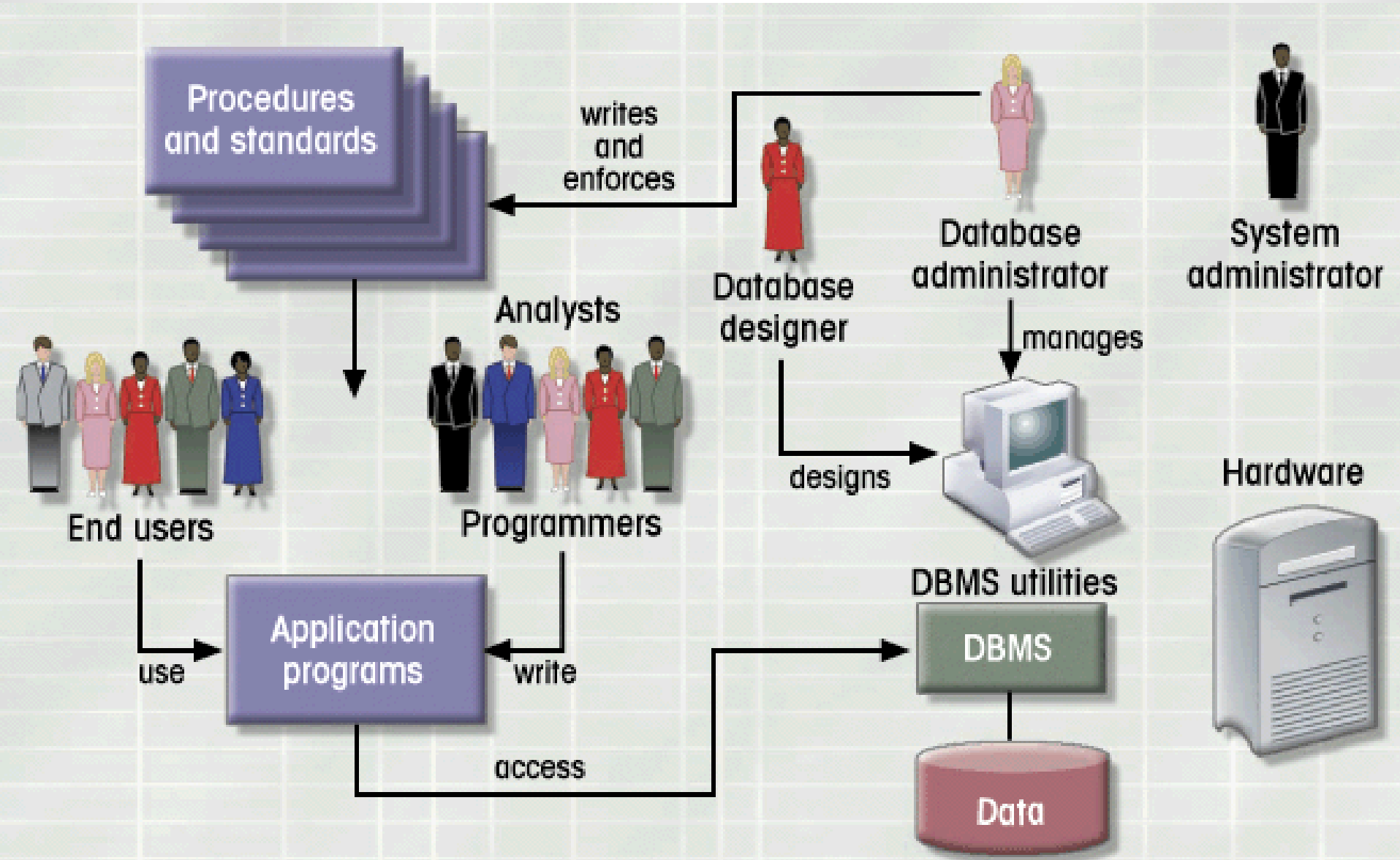
Database Architecture

- **DATA:** Includes database definition, data integrity logic and all other objects that are associated with the data (The role of the DBMS).
- **Business:** Includes the application logic and business rules. For example, ensuring that the employee's salary won't exceed 5000 LE and must be positive and so on.
- **Presentation:** The way of presenting the data to users, that is usually implemented on the client (application program).

Database Architecture (cont.)



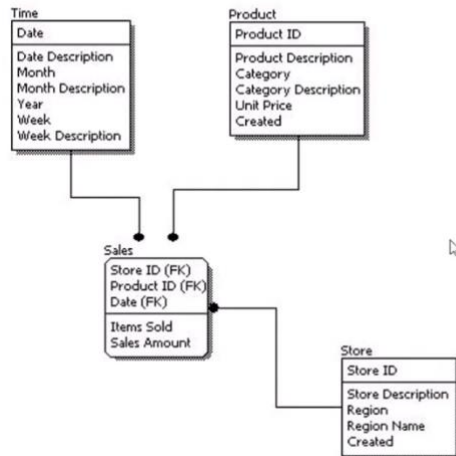
Who deals with a database?



Data Models

➤ Logical data model:

Provides logical representation of the structure of a database.

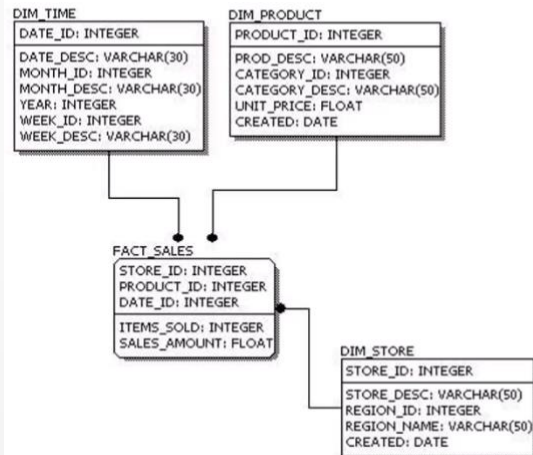


- Presence of Attributes for each Entity
- Key Attributes
- Non-Key Attributes
- Primary Key – Foreign Key Relationships
- User Friendly Attribute names
- More detailed than Conceptual Model
- Database agnostic
- Bit more effort required to enhance, in comparison to Conceptual Model
- Data Modeling tools like ERWin or PowerDesigner can be used to create Logical Data Models. This can be automatically converted to a Physical Data Model with the help of these tools.

Data Models

➤ Physical data model :

Describes how data is stored in the computer and the access path needed to access and search for data.



- Entities referred to as Tables
- Attributes referred to as Columns
- Database compatible table names
- Database compatible column names
- Database specific data types
- Difficult for users to understand
- Significantly more effort required to enhance in comparison to Logical Model
- Will include indexes, constraints, triggers & other DB objects
- Difficult to port to a different database, once design is finalized.
- Tools like ERWin and PowerDesigner can help in automatically porting over the Logical Data Model to Physical Data Models of different versions.

What is SQL?

- SQL stands for Structured Query Language.
- SQL lets you access and manipulate databases.
- SQL is an ANSI (American National Standards Institute) standard.

Why SQL?

- SQL allows users to access data in the relational database management systems
- SQL can create and drop databases
- SQL can create and drop tables in a database
- SQL can insert, select, update, delete records in a database
- SQL allows users to create view, stored procedure and functions in a database.
- SQL can set permissions.

What is Microsoft SQL Server?

- Microsoft SQL Server is a relational database management system, or RDBMS, that supports a wide variety of transaction processing.
- Like other RDBMS software, Microsoft SQL Server is built on top of SQL, a standardized programming language that database administrators (DBAs) and other IT professionals use to manage databases and query the data they contain.

What is Microsoft SQL Server? (cont.)

- The core component of Microsoft SQL Server is the SQL Server Database Engine, which controls data storage, processing and security. It includes a relational engine that processes commands and queries, and a storage engine that manages database files, tables, pages, indexes, data buffers and transactions.
- Stored procedures, triggers, views and other database objects are also created and executed by the Database Engine .

SQL Server Management Studio (SSMS)

- SQL Server Management Studio (SSMS) is an integrated environment for managing any SQL infrastructure.
- Use SSMS to access, configure, manage and develop all components of SQL Server Database.

Microsoft SQL Server Installation

Overview of SQL Server Management Studio

Any Questions?

Thank You...